An extracellular-matrix gene program in clear-cell renal cell carcinoma stroma

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Code, environment and every number and figure: **github.com/FluhrLeviel/opeongo_assignment** (`uv sync && uv run opengo all`). Decisions and their evidence: `outputs/tables/decisions_log.tsv`.

KEY FINDINGS

1. Cohort C was excluded because every cell carries at most 409 non-mitochondrial counts. Cohorts A (single-nucleus, 25 donors) and B (single-cell, 11 donors) were combined with Harmony, which aligned cell identities without erasing tumor/adjacent structure.
2. The ten-gene panel is a stromal program. In the fibroblast + mural compartment, cohort B shows higher expression in tumor than in adjacent tissue in all 9 paired donors (+1.04 SD, 95 % CI 0.63–1.46), and this is not explained by a shift in the fibroblast:pericyte mix. Fibrillar collagens, FN1 and SPARC drive the increase; DCN and LUM do not follow.
3. Cohort A (3 paired donors) is too imprecise to confirm or refute this, so the pooled estimate rests on B and depends on the pooling model.
4. Only 3 donors have matched single-nucleus, bulk RNA and protein data, too few to decide. The whole CPTAC bulk RNA (72 pairs) and protein (80 pairs) cohorts show the same gene-level pattern, which may reflect per-cell change, expansion of mural and vascular tissue, or both.

1 Quality, cohorts and integration

Loading and reconciliation. We restricted cohort A's raw droplet matrices to the 141,950 nuclei annotated by the authors (34 specimens; every annotation matched a barcode). Cohort B (157,881 cells, 26 specimens) and cohort C (29,499 cells, 13 ccRCC or adjacent specimens; the chromophobe specimen is out of scope) were loaded in full. Cell counts per specimen match the package manifest. Genes were matched by Ensembl identifier in A and C and by unique symbol in B, leaving 29,531 shared genes, including all ten panel genes. One suggested label was corrected: the guide maps A's "TRM" nuclei (n = 510) to lymphoid, but they express CSF1R as often as macrophages do (50 %) and almost never express CD3D/E, CD2 or CD8A (≤ 2 %), so we treated them as myeloid.

Quality control. We flagged cells per specimen and per assay rather than with global cut-offs. A cell was flagged if its counts or genes fell more than 5 median absolute deviations below the specimen median, its counts exceeded 5 MAD above it, or Scrublet called it a doublet. It was also flagged if its mitochondrial fraction exceeded the specimen median by 3 MAD and exceeded 5 % in nuclei (where mitochondrial reads indicate ambient RNA) or 25 % in whole kidney cells (where proximal-tubule cells are naturally high). Cohorts A and B had already been filtered by their authors, and B's quality fields match our recomputation exactly. Removing flagged cells cost 4.2 % of A and 0.9 % of B, and no specimen was removed. Removal was uneven across populations: under 4 % in stroma and endothelium, but 22 % of A lymphoid, 10 % of A myeloid and 13 % of B non-malignant epithelial cells (Fig. 1f). This is one reason we never compare population proportions. **Cohort C was excluded.** Every C cell has at most 409 non-mitochondrial counts, whereas A and B reach 10^4–10^5 (Fig. 1a). Mitochondrial reads make up 76 % of C's counts, its two worst specimens are tumors from two of its four paired donors, and DCN and LUM are detected in only 0.2 % of its cells. A marker-based labelling, validated on A and B (81 % and 95 % agreement with author labels; `stage2_marker_label_accuracy.tsv`), still recovers broad populations in C, but per-cell ECM signal there is too sparse to support a comparison.

Baseline versus integration. We embedded a stratified subsample of 87,489 cells (up to 1,000 per specimen plus every stromal and endothelial cell). Cohort A on its own, our starting point, separates cell types well (cell-type ASW 0.63) and shows tumor/adjacent structure within stroma (silhouette 0.26). Adding cohort B without correction separates the cells by cohort (Fig. 1b). In whole-specimen pseudobulk, the first principal component carries 83 % of the variance and is almost entirely cohort (R^2 0.99), i.e. single-nucleus versus single-cell (`stage2_pseudobulk_pca_r2_ALL.tsv`). Harmony with cohort as the batch (Fig. 1c, d) raised cohort mixing within populations (batch ASW 0.49 → 0.81), cell-type separation (0.62 → 0.65) and cross-cohort label transfer (balanced accuracy 0.72 → 0.88), while largely keeping the tumor/adjacent structure in stroma (0.24 → 0.20; Fig. 1e). Using specimen as the batch erased that structure (0.02), because specimen is confounded with condition, so we rejected it.

Integration with batch = cohort *preserved* every major population (kNN purity 0.89–1.00) and the within-population condition structure. It *weakened* the malignant/non-malignant epithelial boundary across cohorts and left global mixing low, because the cohorts differ in composition (A is mostly tumor nuclei, B mostly immune cells) and in assay. Nothing was *lost* with batch = cohort; with batch = specimen the tumor/adjacent structure was lost. Excluding C forfeits its four paired donors. *Combining cohorts adds* two things. After Harmony, A's labels predict 99 % of B's fibroblasts and pericytes as stroma (69 % without integration), which supports treating them as one compartment. And paired donors for that compartment rise from 3 to 12 (Fig. 1g). We used the embedding only for visualisation and label transfer; no statistical test uses it.

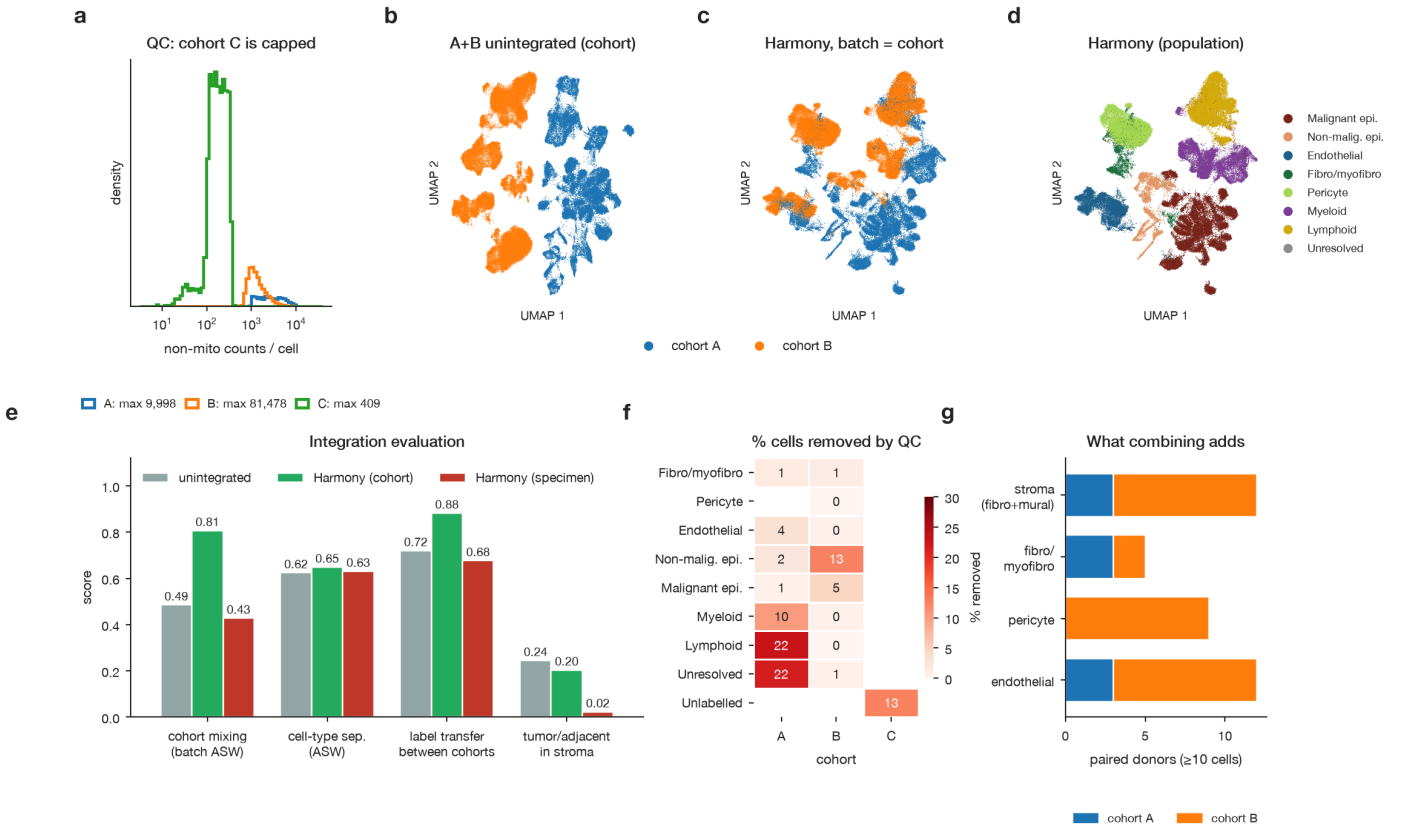


Figure 1 | Quality control and integration. a, Non-mitochondrial counts per cell; cohort C has a hard ceiling. **b–d**, UMAPs of the A+B subsample, unintegrated (b) and after Harmony with batch = cohort, coloured by cohort (c) or population (d). **e**, Integration metrics: batch ASW = 1 − |silhouette of cohort| within each population; cell-type ASW = rescaled silhouette of populations; label transfer = mean A→B and B→A balanced accuracy of a 15-nearest-neighbour classifier; tumor/adjacent = silhouette of condition within stroma (mean of A and B). **f**, Percentage of cells removed by QC per population. **g**, Paired donors with ≥ 10 cells in both conditions per candidate population.

2 The ECM program

Where the program is expressed. In both cohorts every panel gene is expressed most highly in the stromal compartment (Fig. 2a). Endothelial cells express SPARC and FN1 strongly but few collagens and no DCN or LUM, and immune and epithelial cells express little of the panel. Within stroma, DCN and LUM are specific to fibroblasts. They are high in B fibroblasts but near zero in B pericytes and in A's "fibroblasts", which non-panel markers (RGS5, NOTCH3, PDGFRB) identify as largely mural cells (Fig. 2a).

Summary score. We summarise the panel as the **mean of per-gene z-scored log2-CPM**, with z computed within cohort and population. The genes co-vary across stromal pseudobulks (median pairwise ρ 0.48 in A and 0.70 in B), and this score agrees with the first principal component (ρ 1.00 and 0.97) and with log summed CPM (ρ 0.96 and 0.98; `stage4_panel_coherence.tsv`). We therefore use the simplest equal-weight score. Because DCN and LUM do not co-vary with the other genes in B, we always report them separately.

Choice of population. We focus on the **combined fibroblast + mural compartment** (A fibroblasts and myofibroblasts; B fibroblasts and pericytes). It expresses the panel most strongly, non-panel markers and label transfer support it as one compartment, and it has 3 + 9 paired donors with at least 10 cells (Fig. 1g). Fibroblasts alone offer only 5 pairs, 2 of them in B. Endothelium also has 12 pairs, but it expresses mainly SPARC and FN1, so we use it as a contrast. Malignant cells have no adjacent counterpart and were not tested.

Testing. For each specimen we summed the raw counts of its QC-passing compartment cells (pseudobulk) and converted them to log2-CPM, using the library size over all genes. Specimens with fewer than 10 compartment cells were dropped (record: `stage5_specimen_inclusion.tsv`). Donors are the replicates. We averaged repeated tumor regions within each donor and compared each donor's tumor with its own adjacent sample. For each cohort we report the mean paired difference with a 95 % t-interval, the number of donors going up, and an exact sign-flip p (all 2^n sign patterns; with n = 3 the smallest possible p is 0.25). We apply Benjamini–Hochberg correction across genes and pool cohorts by fixed- and random-effects meta-analysis.

Table 1 | Tumor – adjacent difference in the fibroblast + mural compartment (panel score in SD units).

Cohort or pooling	Paired donors	Difference [95 % CI]	Donors up	Sign-flip p
Cohort B (single-cell)	**9**	**+1.04 [+0.63, +1.46]**	**9/9**	**0.004**
Cohort A (single-nucleus)	3	+0.02 [−1.77, +1.81]	1/3	1.0
A + B, fixed effect / pooled sign-flip	12	+0.88 [+0.56, +1.21]	10/12	0.003
A + B, random effects (two cohorts)	12	+0.60 [−0.39, +1.59]		

Tumor versus adjacent. In cohort B the panel score is higher in tumor in all nine donors (Table 1; Fig. 2b). COL1A2, COL3A1, FN1 and SPARC rise in every donor (log2FC +1.7 to +2.6, q = 0.01), and COL1A1, COL6A1 and COL6A2 rise as well (q ≤ 0.03; Fig. 2d). **DCN and LUM do not** (log2FC −1.7 and −0.7, q ≥ 0.17), and they vary by up to ±5 log2 units between donors (Fig. 2d). That is consistent with their coming from the roughly 3 % of B's compartment labelled as fibroblasts. Cohort A shows no consistent direction (Fig. 2c). A mixed model that also uses A's tumor-only donors (24 donors) gives −0.18 [−0.85, +0.48]. Those extra donors contribute only between-donor information, so the contrast still rests on four adjacent specimens (`stage5_mixed_model.tsv`).

Does adding cohorts change the finding? Not decisively. Cohort A's interval contains B's estimate, so A is uninformative rather than contradictory (Table 1). The pooled result depends on the model: the fixed-effect interval excludes zero and the random-effects interval does not. The random-effects heterogeneity (I^2 = 80 %) is not reliable here, because it rests on two cohorts, one with n = 3, each standardised by its own SD.

Robustness check. The main uncertainty is composition: the B effect might reflect more fibroblasts in tumor stroma rather than a change in the cells themselves. If so, it should vanish in pure mural cells and without the fibroblast genes, and the fibroblast share should rise in tumors. None of these happens. The effect holds in pericytes alone (+1.27 [+0.85, +1.70], 9/9 up) and without DCN and LUM (+1.41 [+0.96, +1.86], 9/9 up; Fig. 2e). The fibroblast share barely changes (+0.04 [−0.06, +0.14]) and rises in only 2 of 9 donors (Fig. 2f). The estimate is also stable with at least 50 cells per specimen (+1.07), with one tumor region per donor (+1.04), in a donor random-intercept model on all specimens (+1.11 [+0.85, +1.36]), and when any single donor is left out (+0.93 to +1.15; Fig. 2e). Endothelial cells show no change (+0.13 [−0.26, +0.52]; Fig. 2e), so this is not a tissue-wide shift in ECM transcripts.

Interpretation. In single-cell data, tumor-associated mural and pericyte cells carry more fibrillar collagen, FN1 and SPARC transcript than adjacent ones in the same patients. Higher transcript levels do not by themselves show active matrix remodelling. Nor is it established that the effect holds in single-nucleus data or in fibroblasts: cohort A has three pairs, a different assay, and a compartment of different make-up.

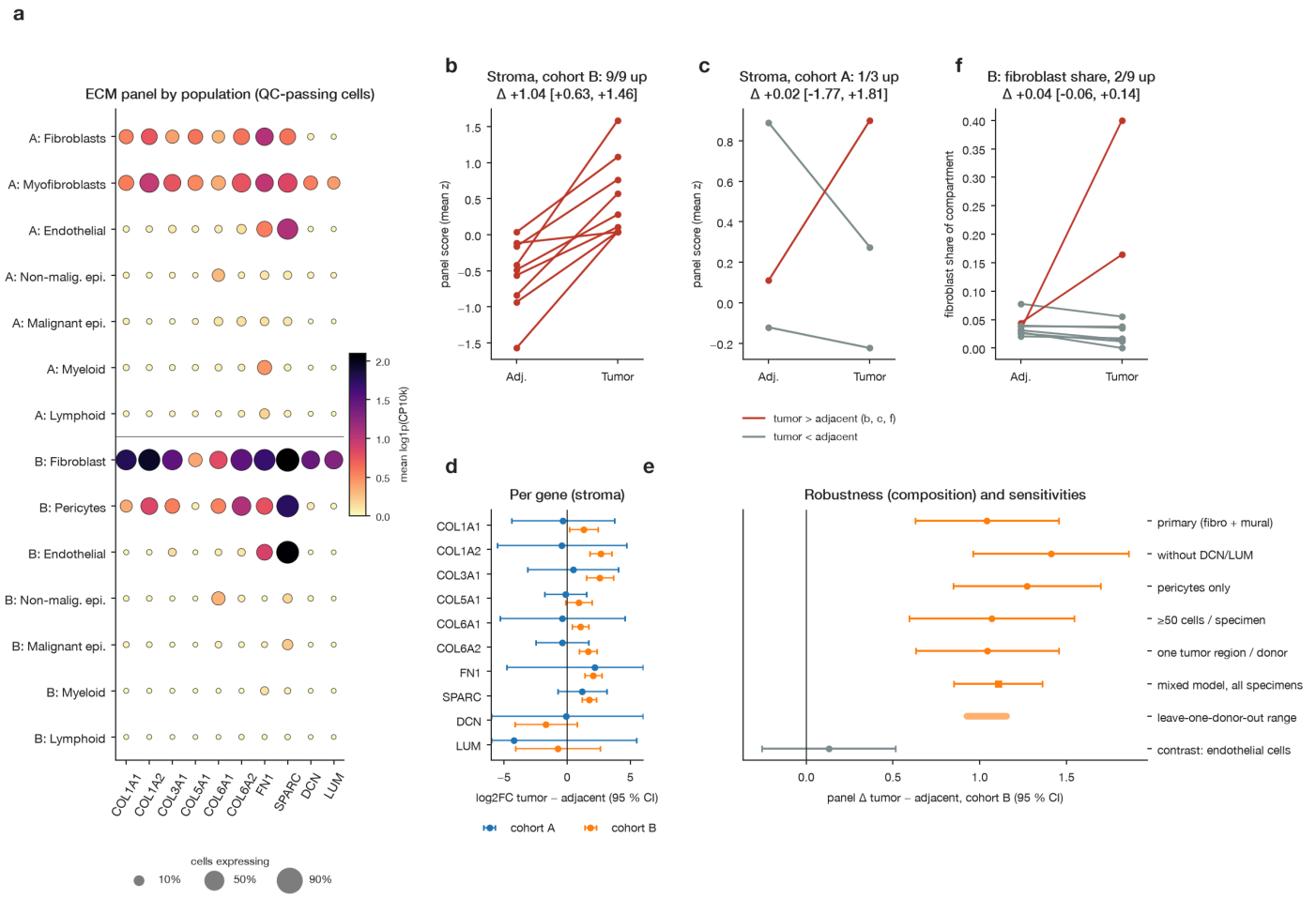


Figure 2 | ECM panel across populations and tumor versus adjacent in stroma. a, Fraction of cells expressing (dot size) and mean expression (colour) per population, QC-passing cells; A and B stroma are split by author label. **b, c**, Donor-level panel score in the fibroblast + mural compartment, adjacent versus tumor, in cohorts B (b) and A (c). **d**, Per-gene paired log2 fold change with 95 % CI (A intervals clipped at ±6). **e**, Cohort B robustness and sensitivity estimates (bar, leave-one-donor-out range; grey, endothelial contrast). **f**, Fibroblast share of B's compartment per donor.

3 Linked bulk RNA and protein

Correspondence. We traced every cohort A specimen through the crosswalk. Of the 34, 29 are linked to bulk data. All 29 resolve through `bulk_samples.tsv` to real columns of the condition-specific RNA and protein files, and every protein aliquot matches the authors' sample table. The only two parent-specimen mismatches are the two flagged links: C3L-00908's protein comes from a different parent specimen, and C3N-00242's from two. That leaves 27 primary links (23 tumor, 4 adjacent). Three donors have primary links for both tumor and adjacent tissue: **C3L-00079, C3L-00088 and C3N-01200**. C3N-00242 serves as a sensitivity case. We did not pair unlinked specimens on donor identity alone (record: `stage6_specimen_inclusion.tsv`). Each assay used a different portion of the parent specimen, and bulk RNA is a donor-level column whose exact aliquot has not been reconciled. Bulk values were used as supplied (log2, normalised); the panel genes have no missing values, although upstream imputation cannot be ruled out.

Observations. In the three matched donors the panel difference is −0.22 for single-nucleus stroma, −0.14 for the whole single-nucleus specimen, +0.48 for bulk RNA and +0.26 for protein (Fig. 3a, d). The stroma value differs from Section 2's +0.02 because here only the linked tumor specimen is used, rather than the average of all tumor regions. The donors disagree and every p is at least 0.5, so the matched subset can neither confirm nor refute the finding. Adding the flagged C3N-00242 link as a sensitivity case moves the matched bulk RNA difference from +0.48 to +0.07 and protein from +0.26 to −0.17 (`stage6_effects.tsv`), which reinforces that conclusion. On average, though, FN1 and SPARC are higher in tumor in every matched assay (Fig. 3a).

For context we used the whole CPTAC cohort. It comes from the same study as cohort A and includes the matched donors, but most of its donors have no single-nucleus data. Tumor panel levels are higher in bulk RNA (+0.52 SD [0.31, 0.73], 54/72 donors) and in protein (+0.68 [0.49, 0.87], 67/80). The gene pattern matches cohort B (Fig. 3a). Collagens rise, FN1 rises in all 72 RNA donors, and SPARC rises in all 72 RNA donors and 78 of 80 protein donors. DCN falls (it rises in only 4 of 72 RNA donors), and LUM falls or stays flat. COL6A1 is the one discrepancy: it falls in bulk RNA but rises in protein and in cohort B.

Composition. Non-panel marker genes in bulk RNA show that tumors lose fibroblast markers (PDGFRA, C7, FBLN1; in 97 % of donors, −1.54 SD) and gain mural (+0.56) and endothelial (+1.63) markers (`stage6_composition_corr.tsv`). Across donors the eight-gene ECM change (without DCN and LUM) tracks fibroblast-marker change (ρ 0.68; Fig. 3b) and mural-marker change (ρ 0.62; Fig. 3c). On average it rises by 0.96 SD despite the fibroblast loss. *Interpretation:* composition strongly shapes bulk differences. More fibroblasts cannot explain the rise, but expansion of mural and endothelial tissue could, alongside or instead of per-cell change, and bulk data cannot separate the two.

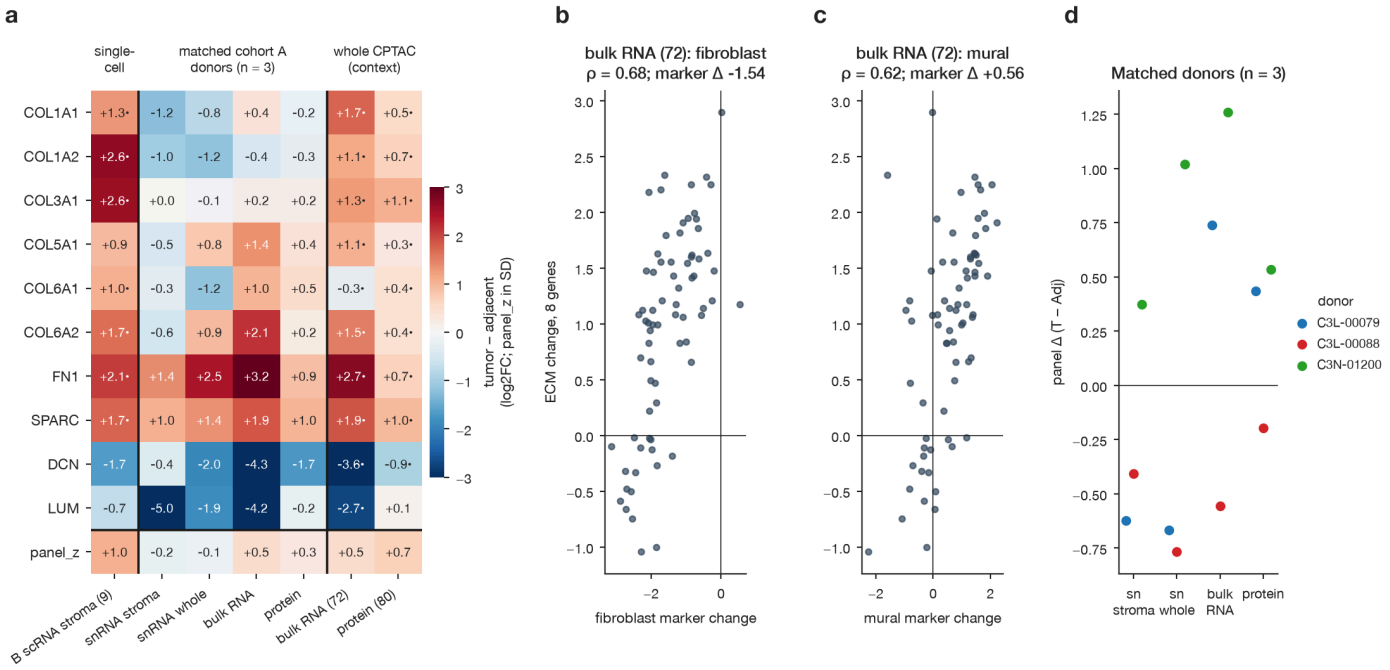


Figure 3 | Linked modalities. a, Tumor − adjacent difference per gene (log2FC) and for the panel score (SD units), each estimated within its own assay. Columns show cohort B stroma (Section 2), the three matched cohort A donors (four assays) and the whole CPTAC bulk RNA and protein cohorts (context, not matched); dots mark BH q < 0.05. **b, c**, Bulk RNA: eight-gene ECM change versus fibroblast (b) and mural (c) marker change per donor. **d**, Panel difference per matched donor in each assay.

4 Agreement, discrepancies and limitations

- **Agreement.** Single-cell cohort B, bulk RNA and protein agree that collagen I/III, FN1 and SPARC are higher in ccRCC than in adjacent kidney, while DCN and LUM are not (Fig. 3a). **Discrepancies.** Single-nucleus cohort A does not reproduce the stromal increase (Fig. 2c), and COL6A1 differs between bulk RNA and protein.
- **Cellular source.** Cell-type-specific RNA does not establish the source of bulk protein. Cohort B places part of the RNA increase in pericytes, but endothelial cells, which expand in tumors, also express FN1 and SPARC strongly (Fig. 2a).
- **Coverage.** Cohort A has 4 adjacent specimens and 3 usable pairs, so no sign-flip p below 0.25 is possible. There are only 3 matched multimodal pairs. About 97 % of B's compartment is labelled as pericytes, so the "stroma" compartment is not identical between cohorts.
- **Composition and capture.** Single-nucleus and single-cell assays capture different cells, bulk differences depend strongly on composition, and the marker-based composition proxy is approximate. Ambient RNA in nuclei may add low-level ECM signal to non-stromal populations.
- **Exclusions.** We excluded cohort C (data integrity), QC-flagged cells, specimens with fewer than 10 compartment cells, two flagged crosswalk links (used for sensitivity only) and, for Section 3, five unlinked single-nucleus specimens. Per-specimen records are in the repository.

Abbreviations. ASW, average silhouette width; kNN, k-nearest neighbours; MAD, median absolute deviation; CPM, counts per million; log2FC, log2 fold change; SD units, per-gene standard deviation of log2-CPM across the cohort's specimens; q, Benjamini–Hochberg-adjusted p; I^2, share of variance between cohorts; mural, pericyte or vascular smooth-muscle-like cells; CPTAC, Clinical Proteomic Tumor Analysis Consortium. Table files cited in the text are in `outputs/tables/`.

1. Heumos L. et al. Best practices for single-cell analysis across modalities. *Nat Rev Genet* (2023).
2. Squair J.W. et al. Confronting false discoveries in single-cell differential expression. *Nat Commun* (2021).
3. Luecken M.D. et al. Benchmarking atlas-level data integration in single-cell genomics. *Nat Methods* (2022).
4. Korsunsky I. et al. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat Methods* (2019).
5. Wu H. et al. Advantages of single-nucleus over single-cell RNA sequencing of adult kidney. *J Am Soc Nephrol* (2019).
6. Clark D.J. et al. Integrated proteogenomic characterization of clear cell renal cell carcinoma. *Cell* (2019).
7. Data: GEO GSE227898 (A), GSE178481 (B), GSE159115 (C); CPTAC ccRCC processed RNA and harmonised protein (package `SOURCES.md`).